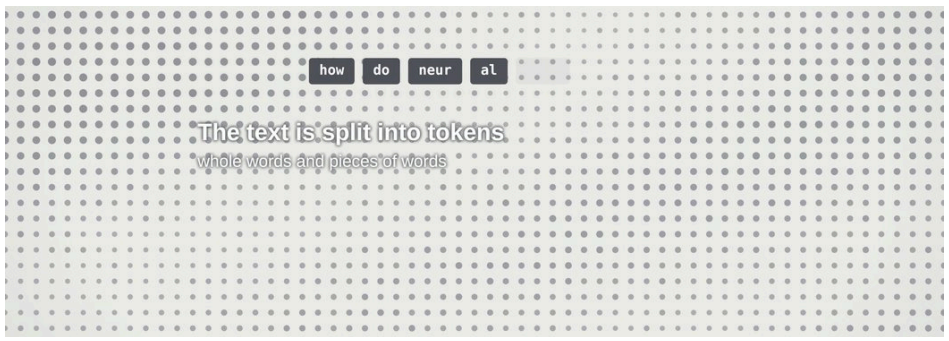
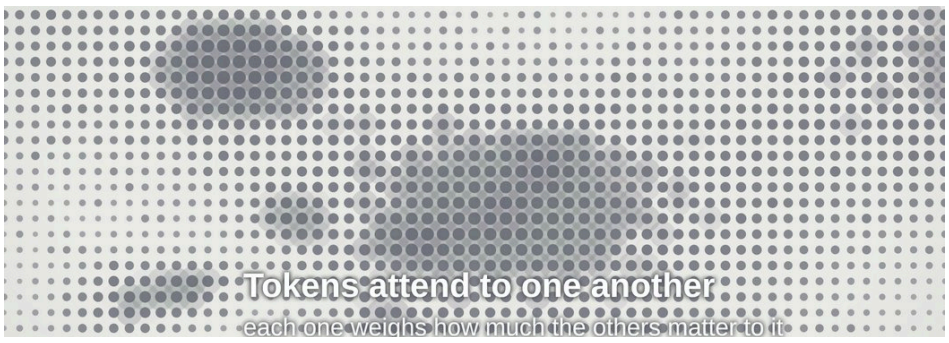
Dot-Matrix Mind: Visualising Language-Model Generation as Overlapping Action

An ambient animation that shows a language model turning a prompt into a reply, drawn in a single material and timed the way an animator times overlapping movement

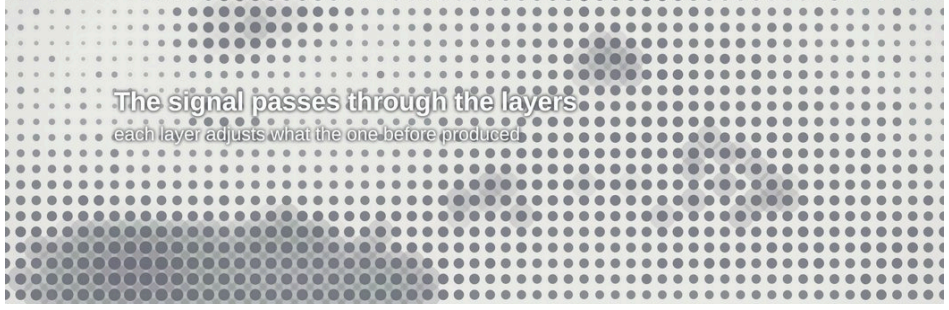
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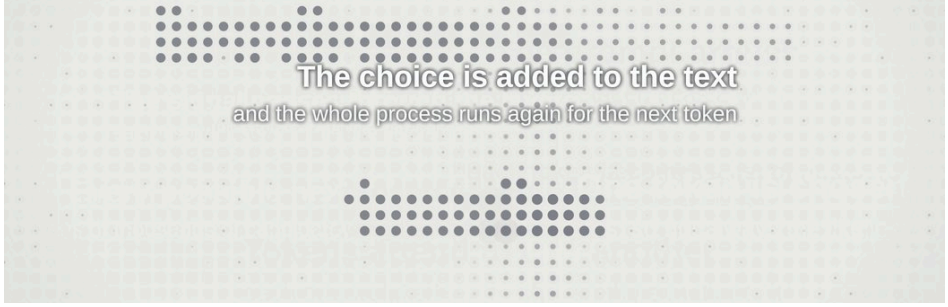
02 · TOKENISATION
THE PROMPT DIVIDED INTO TOKENS



04 · ATTENTION
SIGNALS MOVING BETWEEN TOKENS



05 · LAYER PASSES
WAVES CROSSING THE STACK



08 · FEEDBACK
THE REPLY WRITTEN IN DOTS

Figure 1 — *Four moments from the 48-second loop, captured from the piece. Each stage carries a short caption, and its size and position change from one loop to the next.*

ABSTRACT

Dot-Matrix Mind is an animated visualisation of how a large language model turns a prompt into a reply. Everything on screen is made from one material, a grid of dots, and the stages of generation are shown through the way those dots brighten and move.

There are eight stages: the prompt arriving, tokenisation, embedding, attention, the passes through the model's layers, the scoring of possible next tokens, the choice of one, and that choice being fed back in. They are not played one after another. Each has its own window in the loop, and the windows **overlap**, so attention is still active as the layer passes begin and the candidate tokens start to appear.

The piece is illustrative. It does not run a model, and the prompt, tokens and reply are fixed. It is built for a screen in a studio, corridor or exhibition space, where people will see it in passing, and for the slower look of a crit or a talk.

MOTIVATION

When language models are explained to students, the explanation is usually a block diagram: labelled boxes joined by arrows (Alammar, 2018). These diagrams are accurate, and I use them, but they suggest a production line in which each stage finishes before the next begins. Three-dimensional walkthroughs show the model in motion and in far more detail (Bycroft, 2023). They reward close study, but they ask for more time than a passing viewer has.

Much of my working life in animation has been about timing, and in particular about when one movement starts while another is still settling. Thomas and Johnston (1981) called this **overlapping action**, and it is a large part of why animated movement feels natural. I wanted to see how a language model would look if its stages were timed in the same way.

The visual language comes from dot-matrix and LED displays, the signboards and early screens that built letters from points of light. Because the text and the computation are drawn in the same dots, the prompt can be seen dissolving into the process and the reply forming out of it.

THE EIGHT STAGES

Each stage has a window in the loop. The windows overlap, and a low level of activity runs underneath all of them, so the field never goes completely still. Below each description is a note on how the stage differs from what happens in a real model.

01
PROMPT

A line sweeps across the field from left to right, and the prompt appears behind it, written in dots.

A real model takes in the whole prompt at once. The sweep is there so it can be read.

02
TOKENISATION

The prompt is divided into tokens, shown as labelled tags. Some words are split into pieces.

The splits shown here were chosen by hand to make the idea visible.

03
EMBEDDING

Each token becomes a long list of numbers that places it among related words. The field turns into a slow, drifting pattern.

The drift stands for that space. It is not a map of it.

04
ATTENTION

Points of light travel between nodes as each token works out how much the others matter to it.

The links are redrawn every loop and do not trace real attention weights.

05
LAYER PASSES

Waves cross the field in several directions as the signal moves through the stack of layers, each adjusting what the last produced.

The layers run in order. The overlapping waves show every token being processed together.

06
PROBABILITY

A flickering cloud appears: the model's scores for each token it could produce next.

A real model scores its whole vocabulary, often tens of thousands of tokens.

07
SAMPLING

One point brightens and the rest of the cloud fades. A single token has been chosen.

The choice is weighted by the scores, so the likeliest token does not always win.

08
FEEDBACK

The choice is written out in dots and added to the text, and the next prompt begins to appear. This loop is called autoregression.

In a real model, each new token needs another full pass through the layers.

TIMING THE STAGES

The loop runs for 48 seconds. On every frame, each stage is given a weight between nought and one, which rises and falls smoothly across that stage's window. Neighbouring windows overlap, in the way an arm is still following through while the body has started its next move.

The brightness of each dot is the sum of the weighted stages, plus the background activity. Where several stages are active at once those sums build up, so the total is compressed before it is drawn. This lets overlapping stages add depth to the image without washing it out to white.

The last window overlaps the first, so the next prompt starts to appear while the reply is still being written, and the loop has no visible join. Each time it comes round, the attention links, the positions of the candidate tokens and the placement of the captions are reshuffled. The sequence stays the same, but no two loops look identical.

DRAWING WITH DOTS

The field is an evenly spaced grid that resizes to fit the screen. Each dot has a fixed position and its own small offset in time, which keeps the background activity from pulsing in unison.

The prompt and the reply are drawn as ordinary text off screen, then sampled onto the grid. Thin strokes can fall between dots, so each dot takes the strongest value from the area around it, which pulls the letters onto the grid.

Brightness sets both the size and the opacity of each dot, and the brightest dots are given a soft halo. Each frame is painted over the last at partial opacity, which leaves short trails behind moving light, much like the afterglow on an old phosphor screen.



Figure 2 — *How the piece is built. The prompt is sampled onto the grid; the eight stage functions, each weighted by its window in the loop, are added to the background activity, compressed, and drawn as dot size and opacity.*

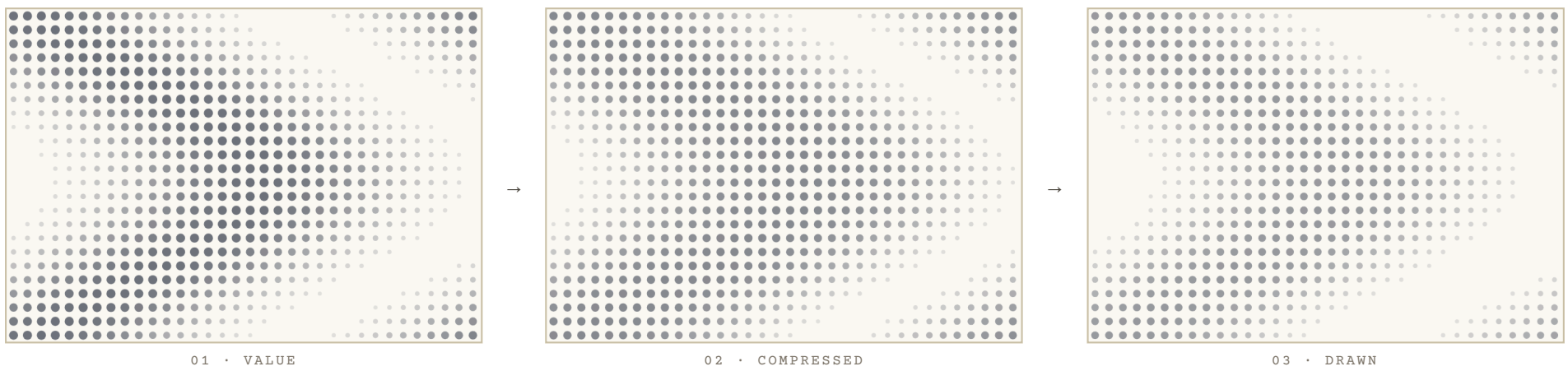


Figure 3 — *From value to dot, shown on a simple test pattern. A stage's raw value (01) is compressed so that overlapping stages do not saturate (02), then drawn as dot size and opacity (03).*

A DIFFERENT PICTURE TO START FROM

In crits and talks, the block diagram tends to fix a picture of AI as a pipeline, with input at one end and output at the other. That picture sits comfortably with the idea that AI is just a tool. Watching generation as one continuous, layered process does not settle what these systems are, but it gives practitioners a different picture to start from.

The piece also works at a distance. A viewer who never reads a caption still sees text dissolve into activity and a reply form out of it, and that is often enough to start a conversation about what happened in between. *Ninety-Two Layers*, its companion piece, is where that conversation can go into detail.

WHAT THE PIECE LEAVES OUT

It is worth being plain about where this is a simplification. Nothing on screen is measured from a model. The attention links, the candidate cloud and the layer waves are drawn to resemble the kind of thing happening at each stage, and they would look the same whatever the prompt. The background flicker is a visual device that keeps the field active between stages; it is not a claim about how a network behaves.

The title needs some care as well. Calling the piece a mind invites the kind of projection that Bender and colleagues warn against (Bender et al., 2021). The piece makes no claim about what, if anything, a model experiences. It shows a sequence of operations, timed so that it can be watched.

FOR PRACTITIONERS

- A picture of generation that can be taken in at a glance, for crits, talks and exhibition
- Each stage named in plain language, with a short caption
- Stages timed as overlapping action, a principle animators already use
- Click to pause on any stage; the L key shows or hides the stage label
- Runs offline from a single file, with no libraries

NEXT STEPS

- A real tokeniser, so that any prompt can be split and shown
- Live replies from a small open model in place of the fixed reply
- A dark version for projection and installation
- Export to looping video, and to still frames for print
- Sessions with practitioners: does an ambient picture of the process change how they talk about working with AI?

REFERENCES & STACK · THOMAS & JOHNSTON (1981) *The Illusion of Life* · VASWANI ET AL. (2017) *Attention Is All You Need* · BENDER ET AL. (2021) · ALAMMAR (2018) *The Illustrated Transformer* · BYCROFT (2023) *LLM Visualization* · LINEAGE: DOT-MATRIX AND LED DISPLAY TYPE · HTML CANVAS 2D · JAVASCRIPT, NO LIBRARIES